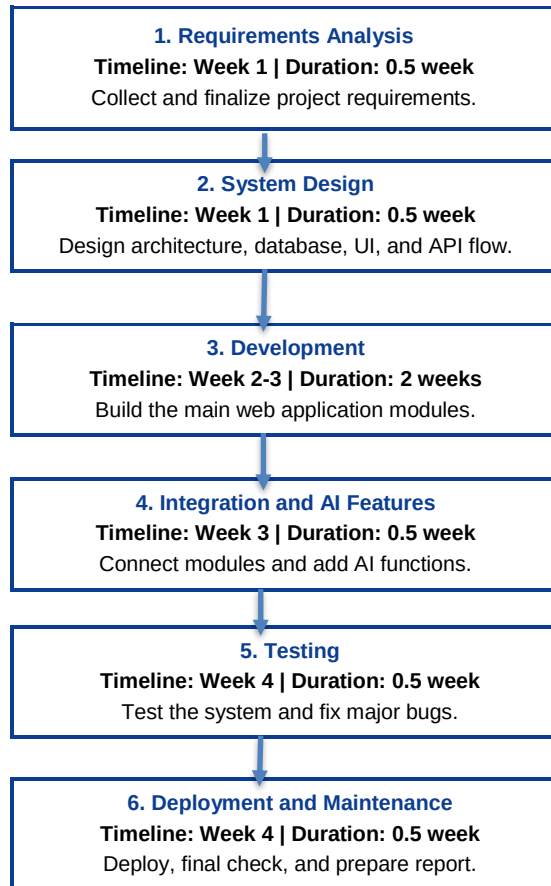


Smart AI-Powered Library Management and Recommendation System

Waterfall Methodology Schedule

Group T | Waterfall Methodology

Team Members: Sangam Poudel, Batbold Chuluunbaatar



Phase	Timeline	Duration	Descriptions	Notes
1. Requirements Analysis	Week 1	0.5 week	Identify project users, define functional and non-functional requirements, confirm AI chatbot, recommendation, smart search, book summary, and admin dashboard requirements.	All members review and agree on requirements before design starts.
2. System Design	Week 1	0.5 week	Design the frontend, backend, database structure, user interface layout, API endpoints, and AI integration approach.	Keep the design simple and realistic.
3. Development	Week 2-3	2 weeks	Develop login/register, user roles, book management, search, borrow/return, database operations, and main web pages.	Tasks can be divided: one member focuses on frontend/UI, the other focuses on backend/database.
4. Integration and AI Features	Week 3	0.5 week	Connect all main modules and add AI features such as chatbot, recommendation engine, AI-generated summaries, and smart search.	Prioritize core AI functions first. Optional AI features can be listed as future enhancements.
5. Testing	Week 4	0.5 week	Test main workflows including login, book search, borrowing, returning, recommendation, chatbot response, and admin dashboard.	Focus on important user flows and fix high-priority bugs first.
6. Deployment and Maintenance	Week 4	0.5 week	Deploy the application, complete final verification, prepare project documentation, and make small improvements after feedback.	Maintenance includes minor bug fixing, final polishing, and preparing future enhancement ideas.